

Counting Schur Rings Over the Cyclic Group of Semiprime Order

Joseph Keller, Andrew Misseldine, Max Sullivan

April 2020

Schur Rings

Let G be a finite group and let $F[G]$ be the group algebra over G with F -coefficients. Let $X \subseteq G$. We will identify X with the element $\sum_{g \in X} g \in F[G]$.

Schur Rings

Let G be a finite group and let $F[G]$ be the group algebra over G with F -coefficients. Let $X \subseteq G$. We will identify X with the element $\sum_{g \in X} g \in F[G]$.

Definition 1.1 (Schur Ring)

Let $C_1, C_2, \dots, C_r$ be a partition (the *primitive sets*) of a finite group G and let $\mathfrak{S}$ be the subspace of $F[G]$ spanned by $C_1, C_2, \dots, C_r$. We say that $\mathfrak{S}$ is a **Schur Ring** over G if

Schur Rings

Let G be a finite group and let $F[G]$ be the group algebra over G with F -coefficients. Let $X \subseteq G$. We will identify X with the element $\sum_{g \in X} g \in F[G]$.

Definition 1.1 (Schur Ring)

Let $C_1, C_2, \dots, C_r$ be a partition (the *primitive sets*) of a finite group G and let $\mathfrak{S}$ be the subspace of $F[G]$ spanned by $C_1, C_2, \dots, C_r$. We say that $\mathfrak{S}$ is a **Schur Ring** over G if

- $C_1 = \{1\}$;
- for each i there is a j such that $C_i^{(-1)} := \{g^{-1} \mid g \in C_i\} = C_j$;
- for each i and j we have $C_i \cdot C_j = \sum_{k=1}^r \lambda_{i,j,k} C_k$, for constants $\lambda_{i,j,k} \in F$.

Schur Rings

Let $\mathfrak{S}$ be a Schur ring over G .

Definition 1.2 ($\mathfrak{S}$ -set)

An $\mathfrak{S}$ -**set** is a union of primitive sets of $\mathfrak{S}$.

Definition 1.3 ($\mathfrak{S}$ -subgroup)

An $\mathfrak{S}$ -set is also a subgroup of G is an $\mathfrak{S}$ -**subgroup**.

Definition 1.4 ($\mathfrak{S}$ -section)

Let K, H be $\mathfrak{S}$ -subgroups. Then we say $[K, H]$ as an $\mathfrak{S}$ -**section** if $1 \leq K \leq H \leq G$ and $K \trianglelefteq G$. We say the section is **proper** if $1 < K \leq H < G$.

Trivial Schur Rings

Definition (Trivial Schur Ring)

A Schur ring is **trivial** if it has only two classes, necessarily 1 and $G \setminus 1$.

For any finite group G , the trivial Schur ring is always available, which we denote as G^0 .

Automorphic Schur Rings

Definition (Automorphic Schur Ring)

Let $\text{Aut}(G)$ be the automorphism group of G , and let $\mathcal{H} \leq \text{Aut}(G)$. Then $\mathcal{H}$ partitions G according to the automorphic action of $\mathcal{H}$ acting on G . This Schur ring is an **automorphic Schur ring** and is denoted $G^{\mathcal{H}}$.

In the special case that $\mathcal{H} = 1$, we simply denote G^1 as G , where all elements are isolated, the **Discrete Schur Ring**.

For abelian groups, let $G^{\pm}$ denote the automorphic Schur rings where elements come in inverse pairs, the **Symmetric Schur Ring**.

Direct Products of Schur Rings

Definition (Direct Product Schur Ring)

Suppose G is a direct product of two groups, say $G = H \times K$. Let $\mathfrak{S}$ and $\mathfrak{T}$ be Schur rings over H and K , respectively. Then we can construct the **direct product** of Schur rings, denoted by $\mathfrak{S} \times \mathfrak{T}$, by the partition

$$\mathfrak{S} \times \mathfrak{T} = \text{Span}\{CD \mid C \in \mathfrak{S}, D \in \mathfrak{T}\}.$$

By construction, H and K are both $\mathfrak{S} \times \mathfrak{T}$ -subgroups.

Schur Subrings

Definition (Schur Subrings)

Let $\mathfrak{S}$ be a Schur rings over G and let H be an $\mathfrak{S}$ -subgroup of G . Then $\mathfrak{S}_H = \mathfrak{S} \cap F[H]$ is a Schur ring over H , called a **Schur Subring**.

Of course, $(\mathfrak{S} \times \mathfrak{T})_H = \mathfrak{S}$ and $(\mathfrak{S} \times \mathfrak{T})_K = \mathfrak{T}$.

Schur Quotient Rings

Definition (Schur Quotient Rings)

Let $\mathfrak{S}$ be a Schur ring over G and let K be a normal $\mathfrak{S}$ -subgroup of G . Let $\varphi : G \rightarrow G/K$ be the natural quotient map. Then $S/K = \varphi(\mathfrak{S})$ is a Schur ring over G/K , called a **Schur Quotient Ring**.

Of course, $(\mathfrak{S} \times \mathfrak{T})/H = \mathfrak{T}$ and $(\mathfrak{S} \times \mathfrak{T})/K = \mathfrak{S}$.

Wedge Products of Schur Rings

Definition (Wedge Decomposable)

A Schur ring $\mathfrak{S}$ is **wedge decomposable** if there is a proper $\mathfrak{S}$ -section $U = [K, H]$ such that for every primitive set C of $\mathfrak{S}$ either $C \subseteq H$ or C is a union of cosets of K . Otherwise, $\mathfrak{S}$ is **wedge-indecomposable**.

Wedge Products of Schur Rings

Definition (Wedge Product Schur Ring)

Let $U = [K, H]$ be a proper section of G , and let $\mathfrak{S}$ and $\mathfrak{T}$ be Schur rings over H and G/K , respectively. Suppose further that

- K is an $\mathfrak{S}$ -subgroup;
- H/K is a $\mathfrak{T}$ -subgroup;
- and $\mathfrak{S}/K = \mathfrak{T}_{H/K}$.

In this case, we define the **wedge product** of $\mathfrak{S}$ and $\mathfrak{T}$, denoted $\mathfrak{S} \wedge_U \mathfrak{T}$, by the rule $\mathfrak{S} \wedge_U \mathfrak{T} = S + \varphi^{-1}(\mathfrak{T})$, where $\varphi : G \rightarrow G/K$ is the natural map.

Traditional Schur Rings

Definition (Traditional Schur Rings)

We say that a Schur ring is **traditional** if it is either trivial, automorphic, direct factorizable, or wedge-decomposable.

Traditional Schur Rings

Definition (Traditional Schur Rings)

We say that a Schur ring is **traditional** if it is either trivial, automorphic, direct factorizable, or wedge-decomposable.

Theorem (The Fundamental Theorem of Schur Rings over Cyclic Groups)

*All Schur rings over cyclic groups are traditional
[Leung and Man, 1996, Leung and Man, 1998].*

Examples

Consider the group $\mathbb{Z}_6$. Then

$\mathbb{Z}_6^0 = \{1, z + z^2 + z^3 + z^4 + z^5\}$ is the trivial Schur ring.

$\mathbb{Z}_6^\pm = \{1, z^1 + z^5, z^2 + z^4, z^3\}$ is the Symmetric Schur ring.

$\mathbb{Z}_2 \wedge \mathbb{Z}_3 = \{1, z^3, z + z^4, z^2 + z^5\}$ is a wedge product Schur ring.

$\mathbb{Z}_2 \times \mathbb{Z}_3^0 = \{1, z^3, z^2 + z^4, z^5 + z\}$ is a direct product Schur ring

Counting Problem

The goal of this research project is to count the number of elementary Schur rings, which means counting each type of Schur ring and using the principles of Inclusion-Exclusion to remove the double counting of the overlap (i.e. an automorphic Schur ring that is also a wedge product).

Counting Problem

The goal of this research project is to count the number of elementary Schur rings, which means counting each type of Schur ring and using the principles of Inclusion-Exclusion to remove the double counting of the overlap (i.e. an automorphic Schur ring that is also a wedge product).

Proposition (1.1)

The number of elementary Schur rings over any group is the number of trivial, automorphic, direct product and wedge product Schur rings, subtract any overlap.

Number of Trivial Schur Rings

For any group $\mathbb{Z}_{pq}$ (or any group at all) there is only one trivial S-ring, so for our counting problem, the trivial S-ring adds one to our final formula.

Number of Trivial Schur Rings

For any group $\mathbb{Z}_{pq}$ (or any group at all) there is only one trivial S-ring, so for our counting problem, the trivial S-ring adds one to our final formula. The trivial S-ring for any group (other than a prime order group) will also not be either automorphic, DP or WP, therefore the intersection is empty and we need not consider it any more. Thus we add the following to our general formula.

Number of Trivial Schur Rings

For any group $\mathbb{Z}_{pq}$ (or any group at all) there is only one trivial S-ring, so for our counting problem, the trivial S-ring adds one to our final formula. The trivial S-ring for any group (other than a prime order group) will also not be either automorphic, DP or WP, therefore the intersection is empty and we need not consider it any more. Thus we add the following to our general formula.

Number of Trivial Schur Rings over $\mathbb{Z}_{pq}$

+1

Number of Automorphic Schur Rings

The number of Automorphic Schur rings over a semiprime group can be counted by counting the subgroups of the automorphic group. This is non-trivial but for some supplemental readings on this see [Călugăreanu, 2004].

Number of Automorphic Schur Rings

The number of Automorphic Schur rings over a semiprime group can be counted by counting the subgroups of the automorphic group. This is non-trivial but for some supplemental readings on this see [Călugăreanu, 2004].

Number of Automorphic Schur Rings of $\text{Aut}(\mathbb{Z}_{p^k})$. (A. Misseldine)

The number of Automorphic Schur Rings is nk , where n is the number of divisors of $p - 1$.

Number of Automorphic Schur Rings

The number of Automorphic Schur rings over a semiprime group can be counted by counting the subgroups of the automorphic group. This is non-trivial but for some supplemental readings on this see [Călugăreanu, 2004].

Number of Automorphic Schur Rings of $\text{Aut}(\mathbb{Z}_{p^k})$. (A. Misseldine)

The number of Automorphic Schur Rings is nk , where n is the number of divisors of $p - 1$.

Combining this with the following result we can begin to enumerate the Automorphic Schur Rings:

$$\text{Aut}(\mathbb{Z}_{pq}) \cong \text{Aut}(\mathbb{Z}_p) \times \text{Aut}(\mathbb{Z}_q) \cong \mathbb{Z}_{p-1} \times \mathbb{Z}_{q-1}$$

Number of Automorphic Schur rings

The number of Automorphic Schur Rings over $\mathbb{Z}_{pq}$ is given by the number of subgroups of the automorphism group of $\mathbb{Z}_{pq}$. This is a counting problem that has to do with the common factors of $p - 1$ and $q - 1$.

Number of Automorphic Schur rings

The number of Automorphic Schur Rings over $\mathbb{Z}_{pq}$ is given by the number of subgroups of the automorphism group of $\mathbb{Z}_{pq}$. This is a counting problem that has to do with the common factors of $p - 1$ and $q - 1$. Therefore the number in general is given by the formula.

Number of Automorphic Schur rings

The number of Automorphic Schur Rings over $\mathbb{Z}_{pq}$ is given by the number of subgroups of the automorphism group of $\mathbb{Z}_{pq}$. This is a counting problem that has to do with the common factors of $p - 1$ and $q - 1$. Therefore the number in general is given by the formula.

Number of Automorphic Schur Rings of $\text{Aut}(\mathbb{Z}_{pq})$

$p = \prod_{i=1}^n r_i^{k_i} + 1$ and $q = \prod_{i=1}^n r_i^{\ell_i} + 1$, where $\{r_1, r_2, \dots, r_n\}$ is a list of distinct primes.

$$\prod_{i=1}^n \left[(k_i + 1)(\ell_i + 1) + \sum_{j=0}^{\min(k_i, \ell_i)} \phi(r_i^{j+1})(k_i - j)(\ell_i - j) \right]$$

Number of Direct Product Schur Rings

For any direct product S-ring, it will be automorphic iff it is a product of automorphic S-rings.

Number of Direct Product Schur Rings

For any direct product S-ring, it will be automorphic iff it is a product of automorphic S-rings.

In fact, since every direct product S-ring over $\mathbb{Z}_{pq}$ will be of the form $\mathcal{Z}_p \times \mathcal{Z}_q$ since p, q are prime, each direct product S-ring will be an automorphic, since the product will combine each of the s-subgroups of each S-ring and create an automorphic s-ring.

Number of Direct Product Schur Rings

For any direct product S-ring, it will be automorphic iff it is a product of automorphic S-rings.

In fact, since every direct product S-ring over $\mathbb{Z}_{pq}$ will be of the form $\mathcal{Z}_p \times \mathcal{Z}_q$ since p, q are prime, each direct product S-ring will be an automorphic, since the product will combine each of the s-subgroups of each S-ring and create an automorphic s-ring. So the set of all direct product S-rings over $\mathbb{Z}_{pq}$ will be a proper subset of the automorphic S-rings, thus we have already counted them.

Number of Wedge Product Schur Rings

Each wedge product S-ring over $\mathbb{Z}_{pq}$ will be of the form $\mathcal{Z}_p \wedge \mathcal{Z}_q$ or $\mathcal{Z}_q \wedge \mathcal{Z}_p$, where $\mathcal{Z}_p, \mathcal{Z}_q$ are S-rings over $\mathbb{Z}_{pq}$. These two forms will be distinct because the wedge product is a non Abelian operation. Thus we will get the following piece of the formula.

Number of Wedge Product Schur Rings

Each wedge product S-ring over $\mathbb{Z}_{pq}$ will be of the form $\mathcal{Z}_p \wedge \mathcal{Z}_q$ or $\mathcal{Z}_q \wedge \mathcal{Z}_p$, where $\mathcal{Z}_p, \mathcal{Z}_q$ are S-rings over $\mathbb{Z}_{pq}$. These two forms will be distinct because the wedge product is a non Abelian operation. Thus we will get the following piece of the formula.

Number of Wedge Product Schur Rings over $\mathbb{Z}_{pq}$

$p = \prod_{i=1}^n r_i^{k_i} + 1$ and $q = \prod_{i=1}^n r_i^{\ell_i} + 1$, where $\{r_1, r_2, \dots, r_n\}$ is a list of distinct primes.

$$2 \prod_{i=1}^n (k_i + 1)(\ell_i + 1)$$

Overlap with Automorphic and Direct Product Schur Rings

Since a wedge product S-ring of the form $\mathcal{Z}_p \wedge \mathcal{Z}_q$ will contain all the s-subgroups of $\mathcal{Z}_p$ but not those of $\mathcal{Z}_q$, since p, q are prime there is no way to get those s-subgroups of $\mathcal{Z}_q$. This rules out any wedge product S-ring from also being an automorphic S-ring over a semiprime order group.

Overlap with Automorphic and Direct Product Schur Rings

Since a wedge product S-ring of the form $\mathcal{Z}_p \wedge \mathcal{Z}_q$ will contain all the s-subgroups of $\mathcal{Z}_p$ but not those of $\mathcal{Z}_q$, since p, q are prime there is no way to get those s-subgroups of $\mathcal{Z}_q$. This rules out any wedge product S-ring from also being an automorphic S-ring over a semiprime order group.

Thus we have counted all 4 types of elementary Schur rings and their overlap with each other. We are now ready for our main result:

Main Result

Number of Schur Rings over a group of Semiprime Order

Let p and q be distinct primes such that $p = \prod_{i=1}^n r_i^{k_i} + 1$ and $q = \prod_{i=1}^n r_i^{\ell_i} + 1$, where $\{r_1, r_2, \dots, r_n\}$ is a list of distinct primes. Then

$$\Omega(pq) = \prod_{i=1}^n \left[(k_i + 1)(\ell_i + 1) + \sum_{j=0}^{\min(k_i, \ell_i)} \phi(r_i^{j+1})(k_i - j)(\ell_i - j) \right]$$

Main Result

Number of Schur Rings over a group of Semiprime Order

Let p and q be distinct primes such that $p = \prod_{i=1}^n r_i^{k_i} + 1$ and $q = \prod_{i=1}^n r_i^{\ell_i} + 1$, where $\{r_1, r_2, \dots, r_n\}$ is a list of distinct primes. Then

$$\Omega(pq) = \prod_{i=1}^n \left[(k_i + 1)(\ell_i + 1) + \sum_{j=0}^{\min(k_i, \ell_i)} \phi(r_i^{j+1})(k_i - j)(\ell_i - j) \right] + 2 \prod_{i=1}^n (k_i + 1)(\ell_i + 1)$$

Main Result

Number of Schur Rings over a group of Semiprime Order

Let p and q be distinct primes such that $p = \prod_{i=1}^n r_i^{k_i} + 1$ and $q = \prod_{i=1}^n r_i^{\ell_i} + 1$, where $\{r_1, r_2, \dots, r_n\}$ is a list of distinct primes. Then

$$\Omega(pq) = \prod_{i=1}^n \left[(k_i + 1)(\ell_i + 1) + \sum_{j=0}^{\min(k_i, \ell_i)} \phi(r_i^{j+1})(k_i - j)(\ell_i - j) \right] \\ + 2 \prod_{i=1}^n (k_i + 1)(\ell_i + 1) + 1,$$

where ϕ denotes Euler's totient function.

Number of Schur rings over Semiprime groups up to 100

Table: Number of Schur Rings over Z_{pq}

n	$\Omega(n)$	n	$\Omega(n)$	n	$\Omega(n)$	n	$\Omega(n)$	n	$\Omega(n)$
6	7	26	19	46	13	65	67	86	25
10	10	33	27	51	35	69	27	87	41
14	13	34	16	55	41	74	28	91	97
15	21	35	41	57	40	77	53	93	53
21	27	38	19	58	19	82	25	94	13
22	13	39	41	62	25	85	60	95	61

Schur Rings over $4p$ Groups

Theorem

For any odd prime $p = 2^k a + 1$ where a is an odd integer and x the number of divisors of $p - 1$

$$\Omega(4p) = \frac{15k + 14}{k + 1}x + 3$$

Further Questions

Questions we still have, opportunity for future research:

Further Questions

Questions we still have, opportunity for future research:

- How many Schur rings are there over groups of order p^2q

Further Questions

Questions we still have, opportunity for future research:

- How many Schur rings are there over groups of order p^2q
- How about groups of order $2pq$?

Further Questions

Questions we still have, opportunity for future research:

- How many Schur rings are there over groups of order p^2q
- How about groups of order $2pq$?
- Some more efficient ways of verifying with a computer our results.

Sources and Supplementary Readings



Călugăreanu, G. (2004).

The total number of subgroups of a finite Abelian group.

Sci. Math. Jpn., 60(1):157–167.



Leung, K. H. and Man, S. H. (1996).

On Schur rings over cyclic groups. II.

J. Algebra, 183(2):273–285.



Leung, K. H. and Man, S. H. (1998).

On Schur rings over cyclic groups.

Isr. J. Math., 106:251–267.

The End

Thank you for listening!